

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
5	(a)	(i)		Radioactive decay/dice throwing is a <u>random</u> process (1) A larger number reduces the effect of random variations [accept anomalous results] / gives more accurate <u>value of half-life</u> / reduces effects of random errors / large sample improves confidence in trend / improves confidence in results (1)	2			2		2
		(ii)		Ticks in 2 nd 3 rd and 5 th boxes i.e. $3 \times (1)$ Note - For each extra tick subtract 1 mark. No negative mark.			3	3		3
		(iii)	I	Represents a non-radioactive material present [accept background [count].			1	1		1
			II	Would always be <u>20</u> higher / would never go below 20			1	1		1
	(b)	(i)		$\frac{56}{8} = 7$ half-lives (1) 68→34→17→8.5→4.25→2.13→1.07→0.54 (1 for halving the mass 7 times ecf) or $\frac{68}{2^7}$ ecf on number of half lives 0.5 [mg] to 1sig fig (1) [Do not accept 1 mg]		3		3	3	
		(ii)		Lead absorbs / blocks [all] the beta [radiation] [emitted from the strontium] (1) Lead <u>reduces</u> the gamma radiation (1) This <u>reduces</u> the exposure / received dose / amount of radiation / harm [to the scientist] (1)	3			3		
				Question 5 total	5	3	5	13	3	7